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Control Valve Capacity Test Procedures

Approved 21 April 2009

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ANSI/ISA-75.02.01-2008 (IEC 60534-2-3 Mod)
Control Valve Capacity Test Procedures

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1 Scope

This test standard utilizes the mathematical equations outlined in ANSI/ISA-75.01.01 (IEC 60534-2-1 Mod)-2007, Flow Equations for Sizing Control Valves, in providing a test procedure for obtaining the following:

a) Valve flow coefficient, C (C_v, K_v)

b) Liquid pressure recovery factors, F_L and F_{LP}

c) Reynolds Number factor, F_R

d) Liquid critical pressure ratio factor, F_F

e) Piping geometry factor, F_P

f) Pressure drop ratio factor, x_T and x_{TP}

g) Valve style modifier, F_d

This standard is intended for industrial process control valves used in flow control of Newtonian fluids. See 4.2 for more information regarding specific valve styles.

2 Purpose

The purpose of this standard is to support ANSI/ISA-75.01.01 (IEC 60534-2-1 Mod)-2007, Flow Equations for Sizing Control Valves, and ANSI/ISA-75.11.01-1985 (R2002), Inherent Flow Characteristic and Rangeability of Control Valves, by providing procedures for testing control valve capacity and related flow coefficients for both compressible and incompressible Newtonian fluids. This standard also provides a procedure to evaluate the major data to calculate the coefficients.

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3 Nomenclature

Symbol description

Symbol	Description	Unit
C	Flow coefficient (C_V, K_V)	Various (see IEC 60534-1) (see note 4)
d	Nominal valve size	mm (in)
D	Internal diameter of the piping	mm (in)
D_1	Internal diameter of upstream piping	mm (in)
D_2	Internal diameter of downstream piping	mm (in)
D_o	Orifice diameter	mm (in)
F_d	Valve style modifier (see Annex A)	Dimensionless (see note 4)
F_F	Liquid critical pressure ratio factor	Dimensionless
F_L	Liquid pressure recovery factor of a control valve without attached fittings	Dimensionless (see note 4)
F_{LP}	Combined liquid pressure recovery factor and piping geometry factor of a control valve with attached fittings	Dimensionless (see note 4)
F_P	Piping geometry factor	Dimensionless
F_R	Reynolds number factor	Dimensionless
F_γ	Specific heat ratio factor	Dimensionless
G_g	Gas specific gravity (ratio of density of flowing gas to density of air with both at standard conditions, which is considered in this practice to be equal to the ratio of the molecular weight of gas to molecular weight of air	Dimensionless
M	Molecular mass of flowing fluid	kg/kg-mol (lb/lb-mol)
N	Numerical constants (see Table 3)	Various (see note 1)
P_1	Inlet absolute static pressure measured at point A (see Figure 1)	kPa or bar (psia) (see note 2)
P_2	Outlet absolute static pressure measured at point B (see Figure 1)	kPa or bar (psia)
P_c	Absolute thermodynamic critical pressure	kPa or bar (psia)
P_V	Absolute vapor pressure of the liquid at inlet temperature	kPa or bar (psia)
ΔP	Differential pressure between upstream and downstream pressure taps ($P_1 - P_2$)	kPa or bar (psi)
Q	Volumetric flow rate (see note 5)	m^3/h (gpm, scfh)
Q_{max}	Maximum flow (choked flow conditions) at given upstream condition	m^3/h (gpm, scfh)
Re_V	Valve Reynolds number	Dimensionless
T_1	Inlet absolute temperature	K (°R)
T_c	Absolute thermodynamic critical temperature	K (°R)
t_s	Absolute reference temperature for standard cubic meter	K (°R)
W	Mass flow rate	kg/h (lbs/h)
x	Ratio of pressure differential to inlet absolute pressure ($\Delta P / P_1$)	Dimensionless
x_T	Pressure differential ratio factor of a control valve without attached fittings at choked flow	Dimensionless (see note 4)

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Symbol	Description	Unit
x_{TP}	Pressure differential ratio factor of a control valve with attached fittings at choked flow	Dimensionless (see note 4)
Y	Expansion factor	Dimensionless
v	Kinematic viscosity	m^2/s (cS) (see note 3)
ρ_1	Density of fluid at P_1 and T_1	kg/m^3 (lb/ft^3)
ρ_1/ρ_o	Relative density (ρ_1/ρ_o = 1.0 for water at 15°C)	Dimensionless
γ	Specific heat ratio	Dimensionless
Subscripts		
1	Upstream conditions	
2	Downstream conditions	

NOTE 1 — To determine the units for the numerical constants, dimensional analysis may be performed on the appropriate equations using the units given in Table 3.

NOTE 2 — 1 bar = 10^2 kPa = 10^5 Pa

NOTE 3 — 1 centistoke = 10^{-6} m^2/s

NOTE 4 — These values are travel-related and shall be stated by the manufacturer.

NOTE 5 — Volumetric flow rates in cubic meters per hour, identified by the symbol *Q*, refer to standard conditions. The standard cubic meter is taken at 1013.25 mbar and 288.6 K (see Table 3).

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4 Test system

4.1 General description

A basic flow test system as shown in Figure 1 includes

a) test specimen;

b) test section;

c) throttling valves;

d) flow-measuring device;

e) pressure taps; and

f) temperature sensor.

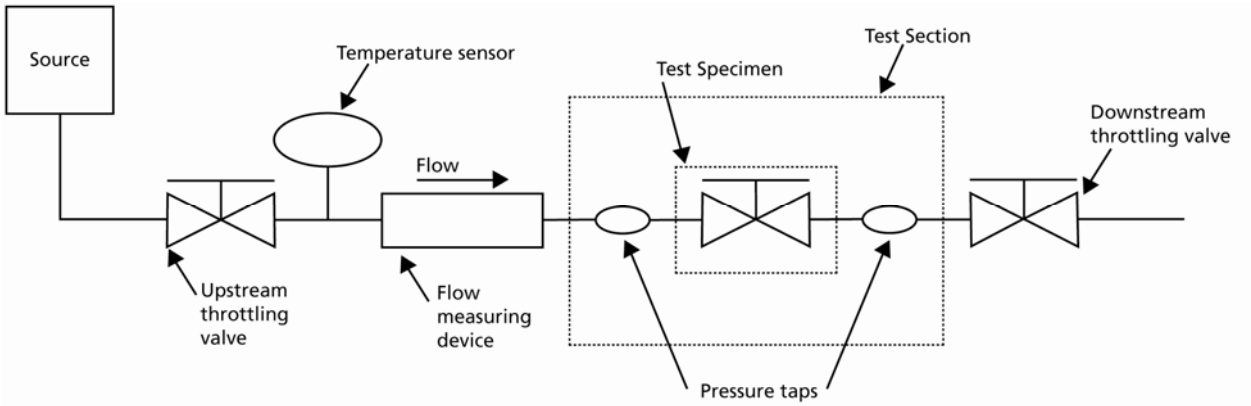


Figure 1 — Basic flow test system

4.2 Test specimen

The test specimen is any valve or combination of valve, pipe reducer, and expander or other devices attached to the valve body for which test data are required. See Annex B for additional examples of test specimens representative of typical field installations.

Additional considerations apply when testing certain styles of control valves. (1) Fractional C (C_v, K_v) valves (valves where $C < 1.00$) require the procedures outlined in Annex D if fully turbulent flow cannot be established because of either high viscosity or low velocities or both. (2) Line-of-sight (e.g., rotary) valves may produce free jets in the downstream test section impacting the location of the pressure recovery zone. See 4.11 for expected accuracies.

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Physical or computer based modeling of control valves as the basis for flow coefficient determination is outside the scope of this standard[1].

4.3 Test section

The upstream and downstream piping adjacent to the test specimen should conform to the nominal size of the test specimen connection and to the length requirements of Figure 2.

The piping on both sides of the test specimen should be Schedule 40 pipe for valves through 250-mm (10-in.) size having a pressure rating up to and including ANSI Class 600. Pipe having 10-mm (0.375-in.) wall may be used for 300-mm (12-in.) through 600-mm (24-in.) sizes. An effort should be made to match the inside diameter at the inlet and outlet of the test specimen with the inside diameter of the adjacent piping for valves outside the above limits.

The inside diameter (D_1, D_2) of the pipe normally should be within ± 2 % of the actual inside diameter of the inlet and outlet of the test specimen for all valve sizes. As the C/d^2 ratio (of the test valve) increases, the mismatch in diameters becomes more problematic. Potential pressure losses associated with the inlet and outlet joints become significant in comparison to the loss associated the valve. Also, as significant discontinuity at the valve outlet could affect the downstream (P_2) pressure measurement. One indication of the significance of mismatched diameters is the value of the piping geometry factor (F_P) based on the internal diameters. This value approaches unity for a standard test, i.e., for equal line and specimen inside diameters. Therefore, to ensure the proper accuracy for the test it shall be demonstrated by either calculation or test that $0.99 \leq F_P \leq 1.01$. If $F_P < 0.99$ it shall be so noted in the test data (see 6.1.5 or 8.1.5). See Annex F for a sample calculation.

The inside surfaces shall be reasonably free of flaking rust or mill scale and without irregularities that could cause excessive fluid frictional losses.

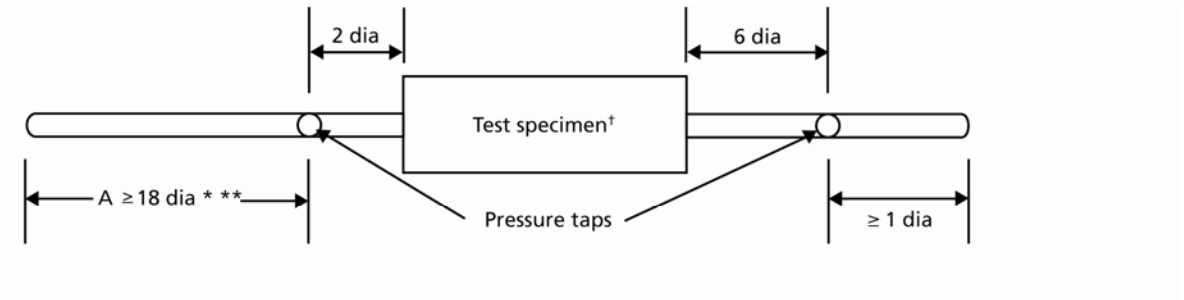
4.4 Throttling valves

The upstream and downstream throttling valves are used to control the pressure differential across the test section pressure taps and to maintain a specific downstream pressure. There are no restrictions as to style of these valves. However, the downstream valve should be of sufficient capacity to ensure that choked flow can be achieved at the test specimen for both compressible and incompressible flow. Vaporization at the upstream throttling valve must be avoided when testing with liquids.

[1] When modeling it is incumbent on the practitioner to utilize sound modeling techniques, to validate the model and scaling relationships to actual flow data, and to document the nature of the model.

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All pipe lengths are in nominal diameters of straight pipe.

* Dimension "A" may be reduced to 8 nominal diameters if straightening vanes are used.

Information concerning the design of straightening vanes can be found in ASME Performance Test Code PTC 19.5-1972, "Applications."

** If an upstream flow disturbance consists of two ells in series and they are in different planes, additional flow conditionining is required. See ASME *Fluid Meters* for additional guidelines for line length.

† See 4.2 for definition of the test specimen.

Figure 2 — Piping requirements, standard test section

4.5 Flow measurement

The flow-measuring instrument may be any device that meets specified accuracy. The accuracy rating of the instrument shall be ± 2 percent of actual output reading. The resolution and repeatability of the instrument shall be within ± 0.5 percent. The measuring instrument shall be calibrated as frequently as necessary to maintain specified accuracy. All guidelines specific to the flow-measuring instrument regarding flow conditioning (e.g., the number of straight pipe diameters, upstream and downstream of the instrument, etc.) shall be followed.

4.6 Pressure taps

Pressure taps shall be provided on the test section piping in accordance with the requirements listed in Figure 2. These pressure taps shall conform to the construction illustrated in Figure 3.

Orientation:

Incompressible fluids — Tap center lines shall be located horizontally to reduce the possibility of air entrapment or dirt collection in the pressure taps.

Compressible fluids — Tap center lines shall be oriented horizontally or vertically above pipe to reduce the possibility of dirt or condensate entrapment.

For butterfly and other rotary valves, the pressure taps shall be aligned (parallel) to the main shaft of the valve to reduce the effect of the velocity head of the flowing fluid on the pressure measurement.

Multiple pressure taps can be used on each test section for averaging pressure measurements. Each tap must conform to the requirements in Figure 3.

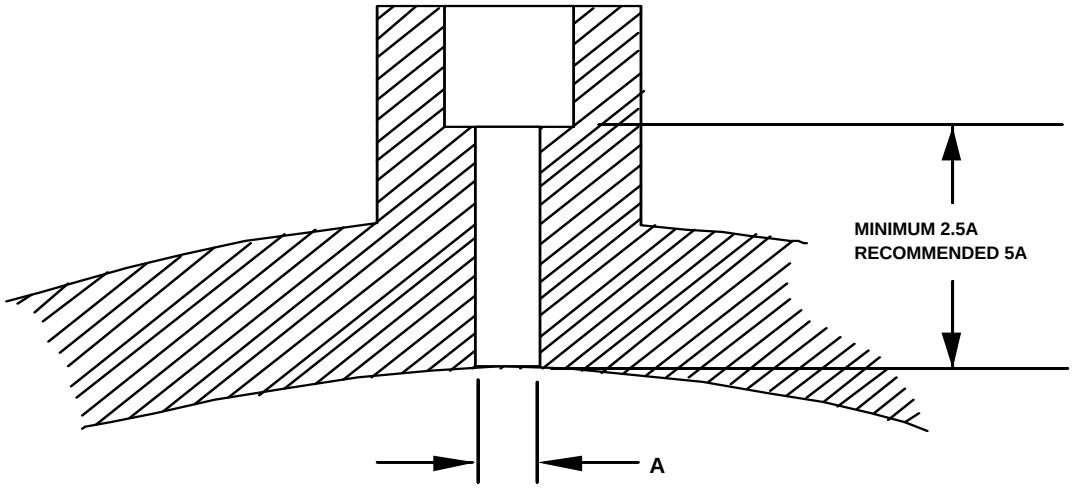
See 4.10 for other installation guidelines.

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	A	A
Size of pipe	**Not exceeding**	**Not less than**
Less than 50 mm (2 in.)	6 mm (1/4 in.)	3 mm (1/8 in.)
50 mm to 75 mm (2 to 3 in.)	9 mm (3/8 in.)	3 mm (1/8 in.)
100 mm to 200 mm (4 to 8 in.)	13 mm (1/2 in.)	3 mm (1/8 in.)
250 mm and greater (10 in. and greater)	19 mm (3/4 in.)	3 mm (1/8 in.)

* Edge of hole must be clean and sharp (i.e., check for corrosion and/or erosion) or slightly rounded, free from burrs, wire edges or other irregularities. In no case shall any fitting protrude inside the pipe.

Any suitable method of making the physical connection is acceptable if above recommendations are adhered to.



Reference: ASME Performance Test Code PTC 19.5-1972, "Applications. Part II of Fluid Meters, Interim Supplement on Instruments and Apparatus."

Figure 3 — Recommended pressure connection

4.7 Pressure measurement

All pressure and pressure differential measurements shall be made using instruments with an accuracy rating of ± 2 percent of actual output reading. Pressure-measuring devices shall be calibrated as frequently as necessary to maintain specified accuracy.

If individual pressure measurements (P_1, P_2) are used in lieu of a single differential pressure measurement (ΔP), care must be taken to select instruments which are accurate enough that the calculated pressure differential value ($P_1 - P_2$) is known with an accuracy at least as good as the accuracy rating stated above for pressure differential measurements.

4.8 Temperature measurement

The fluid temperature shall be measured using an instrument with an accuracy rating of ± 1 °C (± 2 °F) of actual output reading.

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The inlet fluid temperature shall remain constant within ± 3 °C (± 5 °F) during the test run to record data for each specific test point.

4.9 Travel measurement

The accuracy rating of the travel-measuring instrument shall be ± 0.5 percent of rated travel.

4.10 Installation of test specimen

The alignment between the center line of the test section piping and the center line of the inlet and outlet of the test specimen shall be as follows:

Table 1 — Test specimen alignment

Pipe Size	Allowable Misalignment
15 mm thru 25 mm (1/2 in. thru 1 in.)	0.8 mm (1/32 in.)
32 mm thru 150 mm (1-1/4 in. thru 6 in.)	1.6 mm (1/16 in.)
200 mm and larger (8 in. and larger)	1 percent of the diameter

Each gasket shall be positioned so that it does not protrude into the flow stream.

4.11 Accuracy of test

Valves having an $\frac{C}{N_{18}d^2} < 0.047$ at tested travel and $x_T < 0.84$ will have a calculated flow coefficient, C (C_v, K_v) of the test specimen within a tolerance of ± 5 percent. The tolerance for valves that do not meet these criteria may exceed 5%. These accuracy statements apply when fully turbulent flow can be established. See Annex D for further information when this is not the case.

See cautions presented in 4.2.

5 Test fluids

5.1 Incompressible fluids

Fresh water or some other incompressible fluid shall be the basic fluid used in this procedure. Inhibitors may be used to prevent or retard corrosion and to prevent the growth of organic matter. The effect of additives on density or viscosity shall be evaluated by computation using the equations in this standard. The sizing coefficient shall not be affected by more than 0.1 percent. Test fluids other than fresh water may be required for obtaining F_R and F_F. Test fluid temperature range for fresh water should be 5 °C (41 °F) to 40 °C (104 °F).

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5.2 Compressible fluids

Air or some other compressible fluid shall be used as the basic fluid in this test procedure. The test fluid shall fall in the ideal gas behavior range under test conditions, and therefore shall have a ratio of specific heats that falls in the range $1.2 \leq \gamma \leq 1.6$ (cf. Cunningham, Driskell). Vapors that may approach their condensation points at the vena contracta of the specimen are not acceptable as test fluids. Care should be taken to avoid internal icing during the test.

6 Test procedure — incompressible fluids

The following instructions are given for the performance of various tests using incompressible fluids.

The procedures for data evaluation of these tests follow in Clause 7.

6.1 Valve flow coefficient, C, test procedure

The following test procedure is required to obtain test data for the calculation of the flow coefficient C (C_v, K_v) at tested travel. The data evaluation procedure is provided in 7.1.

6.1.1 Install the test specimen without reducers or other attached devices in accordance with piping requirements in Figure 2.

6.1.2 Flow tests shall include flow measurements at three widely spaced pressure differentials within the fully turbulent, non-vaporizing region. The suggested differential pressures are

a) just below the onset of cavitation or the maximum available in the test facility, whichever is less;

b) about 50 percent of the pressure differential of (a); and

c) about 10 percent of the pressure differential of (a) and shall be measured across the test section pressure taps with the valve at the selected travel.

Flow tests should be conducted at a minimum valve Reynolds Number, Re_v, of 100,000 (see Equation 5). If it is not possible to attain a minimum valve Reynolds Number of 100,000, then a compressible flow coefficient test should be considered (also see Annex D). Deviations and reason for the deviations from standard requirements shall be recorded.

Care should be exercised to ensure that the flow rate through the test specimen and the flow measurement device are in fact the same prior to recording data measurements. Compressible flow is potentially problematic. Precautionary steps include establishing steady-state flow through the test system and minimizing the distance between the test specimen and flow measurement device; allow sufficient time after any transient occurring at startup or test valve travel changes.

For large valves where flow source limitations are reached, lower pressure differentials may be used optionally as long as turbulent flow is maintained. Deviations from standard requirements shall be recorded.

6.1.3 In order to keep the downstream portion of the test section filled and to prevent vaporization of the liquid, the absolute upstream pressure shall be maintained at a minimum of $2\Delta P/F_L^2$ or $P_{atm}+2$ psi, whichever is greater. If the liquid pressure recovery factor, F_L, of the test specimen is unknown, a conservative (i.e. low) estimate may be used. See Annex A for typical F_L values. Table 2 provides the minimum upstream pressures for selected values of ΔP and F_L. The line velocity should not exceed 13.7 m/s (45 ft/s) to avoid vaporization in fresh water.

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6.1.4 The valve flow test shall be performed at rated valve travel (normally 100% of available valve travel). Optional tests may be performed at other travels of interest (e.g., 5%, 10%, 20% and every subsequent 10% of rated travel up to and including 100%) or any other desired points to more fully determine the inherent flow characteristic of the specimen (i.e., linear, equal percent, quick opening, etc.).

6.1.5 The following data shall be recorded using the provisions in Clause 4:

a) Valve travel

b) Upstream pressure (P_1)

c) Differential pressure (ΔP) across test section pressure taps

d) Volumetric flow rate (Q) (measurement error not exceeding ± 2 percent of actual value)

e) Fluid inlet temperature (T_1) (measurement error not exceeding ± 1 °C [± 2 °F])

f) Barometric pressure (measurement error not exceeding ± 2 percent of actual value)

g) Physical description of test specimen (i.e., type of valve, flow direction, etc.)

h) Physical description of test system and test fluid

i) Any deviation from the provisions of this standard.

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Table 2 — Minimum upstream test pressure for a temperature range of 5 °C to 40 °C (41 °F to 104 °F)

		Pressure differential used in valve flow coefficient test, ΔP						
	kPa	35	70	100	140	350	700	1400
	bar	0.35	0.70	1.0	1.4	3.5	7.0	14
	psi	5.0	10	15	20	50	100	200
FL		**Minimum absolute upstream pressure, P_1**						
0.5	kPa	280	560	800	1100	2800	5600	11000
	bar	2.8	5.6	8.0	11	28	56	110.0
	psia	40	80	120	160	400	800	1600
0.6	kPa	190	390	560	780	1900	3900	7800
	bar	1.9	3.9	5.6	7.8	19	39	78
	psia	28	56	83	110	280	560	1100
0.7	kPa	140	290	410	570	1400	2900	5700
	bar	1.4	2.9	4.1	5.7	14	29.0	57
	psia	20	41	61	82	200	410	820
0.8	kPa	120*	220	310	440	1100	2200	4400
	bar	1.2*	2.2	3.1	4.4	11.0	22	44
	psia	17*	31	47	63	160	310	630
0.9	kPa	120*	170	250	350	860	1700	3500
	bar	1.2*	1.7	2.5	3.5	8.6	17	35
	psia	17*	25	37	49	120	250	490

* Minimum upstream pressures have been calculated to provide a downstream gage pressure of at least 14 kPa (0.14 bar) (2.0 psig) above atmospheric pressure.

NOTE 1 — Upstream pressures were calculated using P_1 min = $2\Delta P/F_L^2$.

NOTE 2 — Upstream pressures were rounded to 2 significant digits while still maintaining a minimum pressure as specified in note (1).

Example: Estimated F_L for valve is 0.7.
Pressure differential is 70 kPa (0.70 bar;10 psi).
From table: Minimum upstream pressure is 290 kPa (2.9 bar; 41 psia).

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6.2 Liquid pressure recovery factor, F_L,Test procedure

The maximum flow rate, Q_{max} , is required in the calculation of the liquid pressure recovery factor, F_L. For a given upstream pressure, the quantity Q_{max} is defined as that flow rate at which a decrease in downstream pressure will not result in an increase in the flow rate. The test procedure required to determine Q_{max} is included in this subclause. The data evaluation procedure including the calculation of F_L is contained in 7.2. The test for F_L and corresponding C (C_v, K_v) must be conducted at identical valve travels. Hence, the tests for both these factors (Q_{max}, F_L)at any valve travel shall be made while the valve is locked in a fixed position.

6.2.1 Install the test specimen without reducers or other attached devices in accordance with piping requirements in Table 1. A separate test shall be performed for each of the travels identified per 6.1.4. In each test the throttling element shall be positioned and secured at the desired value of travel.

6.2.2 The downstream throttling valve shall be in the fully open position. Then, with a preselected upstream pressure, the flow rate will be measured and the downstream pressure recorded. Table 2 has been provided to assist the user in selecting an upstream pressure. This test establishes a "maximum" pressure differential for the test specimen in this test system.

A second test run shall be made with the pressure differential maintained at 90 percent of the pressure differential determined in the first test with the same upstream pressure. If the flow rate in the second test is within 2 percent of the flow rate in the first test, the "maximum" or choked flow rate has been established. If not, the test procedure must be repeated at a higher upstream pressure. If choked flow cannot be obtained, the published value of F_L must be based on the maximum measurement attainable, with an accompanying notation that the actual value exceeds the published value, e.g., $F_L > 0.87$. See Annex E for a more detailed “long form” procedure.

NOTE — Values of upstream pressure and pressure differential used in this procedure are those values measured at the pressure taps.

6.2.3 The following data shall be recorded using the provisions in Clause 4:

a) Valve travel

b) Upstream pressure (P_1)

c) Differential pressure (ΔP) across test section pressure taps

d) Volumetric flow rate (Q)

e) Fluid temperature

f) Barometric pressure

g) Physical description of test specimen (i.e., type of valve, flow direction, etc.)

h) Physical description of test system and test fluid

i) Any deviation from the provisions of this standard

6.3 Piping geometry factor, F_P, test procedure

The piping geometry factor, F_P , modifies the valve sizing coefficient for reducers or other devices attached to the valve body that are not in accord with the test section. It is the ratio of the installed C (C_v, K_v) with these reducers or other devices attached to the valve body to the rated C (C_v, K_v) of the

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valve installed in a standard test section and tested under identical service conditions. This factor is obtained by replacing the valve with the desired combination of valve, reducers, and/or other devices and then conducting the flow test outlined in 6.1, treating the combination of the valve and reducers as the test specimen for the purpose of determining test section line size. For example, a 100-mm (4-in.) valve between reducers in a 150-mm (6-in.) line would use pressure tap locations based on 150-mm (6-in.) nominal diameter. The data evaluation procedure is provided in 7.3.

6.4 Combination (product) of liquid pressure recovery factor F_L and piping geometry factor F_P, F_{LP} test procedure

Perform the tests outlined for F_L in 6.2, replacing the valve with the desired combination of valve and pipe reducers or other devices and treating the combination of valve and reducers as the test specimen. The data evaluation procedure is provided in 7.4.

6.5 Reynolds Number factor, F_R, test procedure

To produce values of the Reynolds Number factor, F_R, nonturbulent flow conditions must be established through the test valve. Such conditions will require low pressure differentials, high viscosity fluids, small values of C (C_v, K_v) or some combination of these. With the exception of valves with very small values of C (C_v, K_v) turbulent flow will always exist when flowing tests are performed in accordance with the procedure outlined in 5.1, and F_R under these conditions will have the value of 1.0.

Determine values of F_R by performing flowing tests with the valve installed in the standard test section without reducers or other devices attached. These tests shall follow the procedure for C (C_v, K_v) determination except that

a) test pressure differentials may be any appropriate values provided that no vaporization of the test fluid occurs within the test valve;

b) minimum upstream test pressure values shown in Table 2 may not apply if the test fluid is not fresh water at 20 °C ± 14 °C (68 °F ± 25 °F); and

c) the test fluid shall be a Newtonian fluid having a viscosity considerably greater than water unless instrumentation is available for accurately measuring very low pressure differentials.

Perform a sufficient number of these tests at each selected valve travel by varying the pressure differential across the valve so that the entire range of conditions, from turbulent to laminar flow, is spanned. The data evaluation procedure is provided in 7.5.

6.6 Liquid critical pressure ratio factor, F_F, test procedures

The liquid critical pressure ratio factor, F_F, is ideally a property of the fluid and its temperature. It is the ratio of the apparent vena contracta pressure at choked flow conditions to the vapor pressure of liquid at inlet temperature.

The quantity of F_F may be determined experimentally, although it is not possible to evaluate F_F, C and F_L concurrently. A test specimen for which F_L and C (C_v, K_v) have been previously established by test in a system utilizing known fluid properties is required. The standard test section without reducers or other devices attached will be used with the test specimen installed. The test procedure outlined in 6.2 for obtaining Q_{max} will be used with the fluid of interest as the test fluid. The data evaluation procedure is in 7.6.

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7 Data evaluation procedure — incompressible fluids

The following procedures are to be used for the evaluation of the data obtained using the test procedures in Clause 6. The pressure differentials used to calculate the flow coefficients and other flow factors were obtained using the test section defined in Table 1. These pressure measurements were made at the pressure taps and include the test section piping between the taps as well as the test specimen.

7.1 C Calculation

7.1.1 Using the data obtained in 6.1, calculate C (C_v, K_v) for each test point at a given valve travel using the equation

(Eq. 1)
$$C = \frac{Q}{N_1}\sqrt{\frac{\rho_1 / \rho_o}{\Delta P}}$$

Round off the calculated value to no more than three significant digits.

7.1.2 The flow coefficient C (C_v, K_v) of the valve is the arithmetic average of the calculated values at each travel tested as obtained from the test data in 6.1.5. The individual values used in computing the average value should fall within ± 2.5% of the average value. The "rated *C*" is the flow coefficient at 100% rated travel.

7.2 F_L Calculation

Calculate F_L as follows:

(Eq. 2)
$$F_L = \frac{Q_{max}}{N_1 C \sqrt{\frac{(P_1 - F_F P_v)}{\rho_1 / \rho_o}}}$$

where P_1 is the pressure at the upstream pressure tap for the Q_{max} determination (see 6.2). If fresh water at 5 to 40 °C (41 to 104 °F) is used F_F has a value of 0.96. If fresh water is not used, F_F for that fluid shall be used[2].

7.3 F_P Calculation

Calculate F_P as follows:

(Eq. 3)
$$F_P = \frac{Q}{N_1 C \sqrt{\frac{\Delta P}{\rho_1 / \rho_o}}}$$

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[2] If the test fluid is a single component fluid it is permissible to use $F_F = 0.96 - 0.28\sqrt{\frac{P_v}{P_c}}$.

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7.4 F_{LP} Calculation

Calculate F_{LP} as follows:

(Eq. 4)
$$F_{LP} = \frac{Q_{max}}{N_1 C \sqrt{\frac{P_1 - F_F P_v}{\rho_1 / \rho_o}}}$$

where P_1 is the pressure at the upstream pressure tap for the Q_{max} determination (see 6.2).

7.5 F_R Calculation

Use test data, obtained as described under 6.5 and in Equation (1) in 7.1 to obtain values of an apparent C (C_v , K_v). This apparent C (C_v , K_v) is equivalent to $F_R C_v$. Therefore, F_R is obtained by dividing the apparent C (C_v , K_v) by the experimental value of C (C_v , K_v) determined for the test valve under standard conditions at the same valve travel. Although data may be correlated in any manner suitable to the experimenter, a method that has proven to provide satisfactory correlations involves the use of the valve Reynolds Number, which may be calculated from

(Eq. 5)
$$\mathrm{Re}_v = \frac{N_4 F_d Q}{\nu \sqrt{F_L C}} \left(\frac{F_L^2 C^2}{N_2 D^4} + 1 \right)^{1/4}$$

where

F_d = valve style modifier, accounts for the effect of geometry on Reynolds Number (see Annex C for additional discussion).

v = kinematic viscosity in centistokes.

Plotting values of F_R versus Re_v will result in the curve that appears as Figure 3 a & b in ANSI/ISA-75.01.01 (IEC 60534-2-1 Mod)-2007, Flow Equations for Sizing Control Valves.

7.6 F_F Calculation

Using the data obtained in accord with 6.6 calculate F_F as follows:

(Eq. 6)
$$F_F = \frac{1}{P_v} \left[P_1 - \frac{\rho_1}{\rho_o} \left(\frac{Q_{\max}}{N_1 F_L C} \right)^2 \right]$$

where P_v is the fluid vapor pressure at the inlet temperature.

8 Test procedure — compressible fluids

The following instructions are given for the performance of various tests using compressible fluids.

The procedures for data evaluation of these tests follow in Clause 9.

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8.1 C Test procedure

The determination of the flow coefficient, C (C_v, K_v) requires flow tests using the following procedure to obtain the necessary test data. The data evaluation procedure is in 9.1. An alternative procedure for calculating C (C_v, K_v) is provided in 8.3.

8.1.1 Install the test specimen without reducers or other devices in accordance with the piping requirements in Table 1.

8.1.2 Flow tests will include flow measurements at three pressure differentials. In order to approach flowing conditions that can be assumed to be incompressible, the pressure drop ratio ($x = \Delta P/P_1$) shall be ≤ 0.02. It is also necessary to ensure that the flowing conditions are operating n the fully turbulent flow regime. A minimum valve Reynolds Number of 100,000 should be established for all test conditions (see Equation 5). Note that actual volumetric flow rate should be used in computing the Reynolds Number.

8.1.3 The valve flow test shall be performed at 100 percent of rated valve travel. Optional tests may be performed at 5 percent and each 10 percent of rated valve travel or any other points of interest to more fully determine the inherent characteristic of the specimen.

8.1.4 The following data shall be recorded using the provisions in Clause 4:

a) Valve travel

b) Upstream pressure (P_1)

c) Differential pressure (ΔP) across test section pressure taps

d) Volumetric flow rate (Q)

e) Fluid temperature (T_1) upstream of valve

f) Barometric pressure

g) Physical description of test specimen (e.g., type of valve, flow direction, etc.)

h) Physical description of test system and test fluid

i) Any deviation from the provisions of this standard.

8.2 x_T Test procedure

The maximum flow rate, Q_{max} , (referred to as choked flow) is required in the calculation of x_T , the pressure drop ratio factor. This factor is the terminal ratio of the differential pressure to absolute upstream pressure ($\Delta P / P_1$), for a given test specimen installed without reducers or other devices. The maximum flow rate is defined as that flow rate at which, for a given upstream pressure, a decrease in downstream pressure will not produce an increase in flow rate. The test procedure required to obtain Q_{max} is contained in this subclause with the data evaluation procedure in 9.2. An alternative procedure for determining x_T is provided in 8.3.

8.2.1 Install the test specimen without reducers or other attached devices in accordance with piping requirements in Table 1. The test specimen shall be at 100 percent of rated travel. Optional tests may be done at other valve travels to more fully understand the possible variation of x_T with valve travel.

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8.2.2 Any upstream supply pressure sufficient to produce choked flow is acceptable, as is any resulting pressure differential across the valve, provided that the criteria for determination of choked flow specified in 8.2.3 are met.

8.2.3 The downstream throttling valve will be in the wide-open position. Then, with a preselected upstream pressure, the flow rate will be measured and the downstream pressure recorded. This test establishes the maximum pressure differential for the test specimen in this test system. A second test shall be conducted using the downstream throttling valve to reduce the pressure differential by 10 percent of the pressure differential determined in the first test (with the same upstream pressure). If the flow rate of this second test is within 0.5 percent of the flow rate for the first test, then the maximum flow rate has been established.

In order to attain the prescribed accuracy, the flow rate instrument accuracy and repeatability requirements of 4.5 must be followed. This series of tests must be made consecutively, using the same instruments, and without alteration to the test setup.

8.2.4 The following data shall be recorded using the provisions in Clause 4:

a) Valve travel

b) Upstream pressure (P_1)

c) Differential pressure (ΔP) across test section pressure taps

d) Volumetric flow rate (Q)

e) Fluid temperature upstream (T_1) of valve

f) Barometric pressure

g) Physical description of test specimen (e.g., type of valve, flow direction, etc.)

h) Physical description of test system and test fluid

i) Any deviation from the provisions of this standard.

8.3 Alternative test procedure for C and x_T

8.3.1 Install the test specimen without reducers or other attached devices in accordance with piping requirements in Table 1. The test specimen shall be at 100 percent of rated travel (or at any other travel of interest).

8.3.2 With a preselected upstream pressure, P_1 ,measurements shall be made of flow rate, Q, upstream fluid temperature, T_1 , differential pressure, ΔP , for a minimum of five well-spaced values of x (the ratio of pressure differential to absolute upstream pressure).

8.3.3 From these data points calculate values of the product YC using the equation:

(Eq. 7)
$$YC = \frac{Q}{N_7 P_1}\sqrt{\frac{G_g T_1}{x}}$$

where Y is the expansion factor defined by

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(Eq. 8) $$Y = 1 - \frac{x}{3F_\gamma x_T}$$

where

(Eq. 9) $$F_\gamma = \frac{\gamma}{1.4}$$

8.3.4 The test points shall be plotted on linear coordinates as (YC) vs. x and a linear curve fitted to the data. If any point deviates by more than 5 percent from the curve, additional test data shall be taken to ascertain if the specimen truly exhibits anomalous behavior.

8.3.5 At least one test point $(YC)_1$ shall fulfill the requirement that

$$(YC)_1 \geq 0.97(YC)_o$$

where $(YC)_o$ corresponds to $x \cong 0$.

8.3.6 At least one test point, $(YC)_n$ shall fulfill the requirement that

$$(YC)_n \leq 0.83\ (YC)_o$$

8.3.7 The value of C (C_v, K_v) for the specimen shall be taken from the curve at x = 0, Y = 1.

The value of x_T for the specimen shall be taken from the curve at YC = 0.667C .

8.4 Piping geometry factor, F_P, test procedure

The piping geometry factor, F_P , modifies the valve sizing coefficient for reducers or other devices attached to the valve body that are not in accord with the test section. The factor F_P is the ratio of the installed C (C_v, K_v) with the reducers or other devices attached to the valve body to the rated C (C_v, K_v) of the valve installed in a standard test section and tested under identical service conditions. This factor is obtained by replacing the valve with the desired combination of valve, reducers, and/or other devices and then conducting the flow test outlined in 8.1, treating the combination of valve and reducers as the test specimen for the purpose of determining test section line size. For example, a 100-mm (4-inch) valve between reducers in a 150-mm (6-inch) line would use pressure tap locations based on a 150-mm (6-inch) nominal diameter. The data evaluation procedure is provided in 9.3.

8.5 x_{TP} Test procedure

Perform the tests outlined for x_T in 8.2, replacing the valve with the desired combination of valve and pipe reducers or other devices and treating the combination of valve and reducers as the test specimen. The data evaluation procedure is provided in 9.4.

9 Data evaluation procedure — compressible fluids

The following procedures shall be used for the evaluation of the data obtained using the test procedures in Clause 7. The pressure differentials used to calculate the flow coefficients and other flow factors shall have been obtained using the test section defined in Table 1. These pressure measurements shall have been made at the pressure taps and include the test section piping between the taps as well as the test specimen.

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9.1 C Calculation

Using the data obtained in 8.1 and assuming the expansion factor Y = 1.0, calculate the flow coefficient, C (C_v, K_v) for each test point using

(Eq. 10) $$C = \frac{Q}{N_7 P_1}\sqrt{\frac{T_1 G_g}{x}}$$

Calculate the arithmetic average of the three test values obtained at rated travel to obtain the rated C (C_v, K_v).

9.2 x_T Calculation

Calculate x_T as follows:

From Equation (7),

(Eq. 11) $$Q = N_7 Y C P_1 \sqrt{\frac{x}{T_1 G_g}}$$

When $x = F_\gamma\ x_T$, then $Q = Q_{max}$

(Eq. 12) $$Q_{\max} = N_7 Y C P_1 \sqrt{\frac{F_\gamma x_T}{T_1 G_g}}$$

rearranging yields

(Eq. 13) $$x_T = \left(\frac{Q_{\max}}{N_7 Y C P_1}\right)^2 \frac{G_g T_1}{F_\gamma}$$

Assuming air as test fluid and substituting Y = 0.667, G_g = 1.0, and F_γ = 1.0:

(Eq. 14) $$x_T = \left(\frac{Q_{\max}}{0.667 N_7 C P_1}\right)^2 T_1$$

Best accuracy is achieved when the instantaneous values of P_1 and T_1 associated with the Q_{max} value are used in Equation 14.

9.3 F_P Calculation

Calculate F_P at rated valve travel (or any other travel being investigated):

(Eq. 15) $$F_P = \frac{Q}{N_7 P_1 C_{rated}\sqrt{\frac{x}{T_1 G_g}}}$$

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9.4 x_{TP} Calculation

Calculate x_{TP} as follows:

From Equation (7),

(Eq. 16)
$$Q = N_7 F_P Y C P_1 \sqrt{\frac{x_{TP}}{G_g T_1}}$$

with F_P added to account for reducers and other devices. When $x = x_{TP}$, $Q = Q_{max}$

(Eq. 17)
$$Q_{\max} = N_7 F_P Y C P_1 \left(\frac{x_{TP}}{G_g T_1} \right)$$

Assuming air as the test fluid:

Y = 0.667

G_g = 1.0

F_γ = 1.0

(Eq. 18)
$$x_{TP} = \left(\frac{Q_{\max}}{0.667 N_7 F_P C P_1} \right)^2 T_1$$

10 Numerical constants

The numerical constants, N, depend on the measurement units used in the general sizing equations. Values for N are listed in Table 3.

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Table 3 — Numerical constants

Constant	Flow Coefficient C		Formulae Units					
	K_v	C_v	$Q^{(1)}$	$P, \Delta P, P_v^{(2)}$	$\rho^{(3)}$	T	d, D	$\nu^{(4)}$
N_1	1.00×10^{-1} 1.00	8.65×10^{-2} 8.65×10^{-1} 1.00	m^3/h m^3/h gpm	kPa bar psia	kg/m^3 kg/m^3 lbm/ft^3	— — —	— — —	— — —
N_2	1.60×10^{-3}	2.14×10^{-3} 8.90×10^{2}	— —	— —	— —	— —	mm in	— —
N_4	7.07×10^{-2}	7.60×10^{-2} 1.73×10^{4}	m^3/h gpm	— —	— —	— —	— —	m^2/s cS
N_7	4.82 4.82×10^{2}	4.17 4.17×10^{2} 1.36×10^{3}	m^3/h m^3/h scfh	kPa bar psia	— — —	— — —	— — —	— — —
N_{18}	8.65×10^{-1}	1.00 6.45×10^{2}	— —	— —	— —	— —	mm in	— —

(1) The standard cubic foot is taken at 14.70 psia and 60 °F and the standard cubic meter at 1013.25 mbar and 288.6 K.

(2) All pressures and temperatures are absolute.

(3) Constant N_1 is technically independent of the density units. However, density units have been shown to help ensure that consistent density units are employed in both the numerator and denominator of density ratios (c.f. Equation 1).

(4) Centistoke = $10^{-6} m^2/sec$

For nomenclature symbol definition, see Clause 3.

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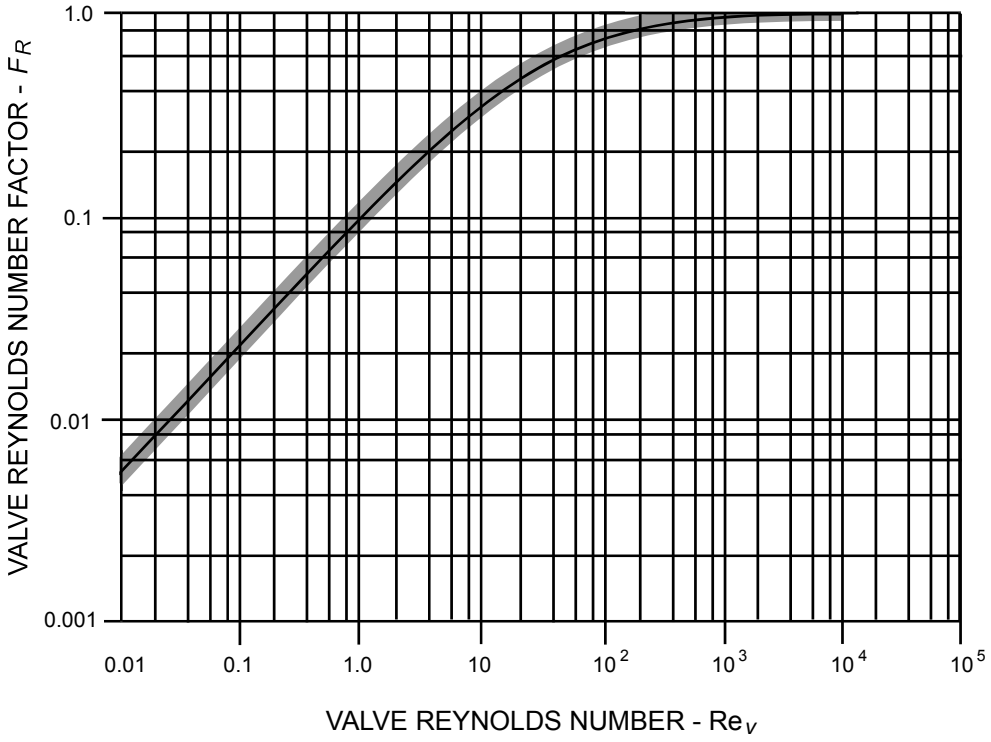


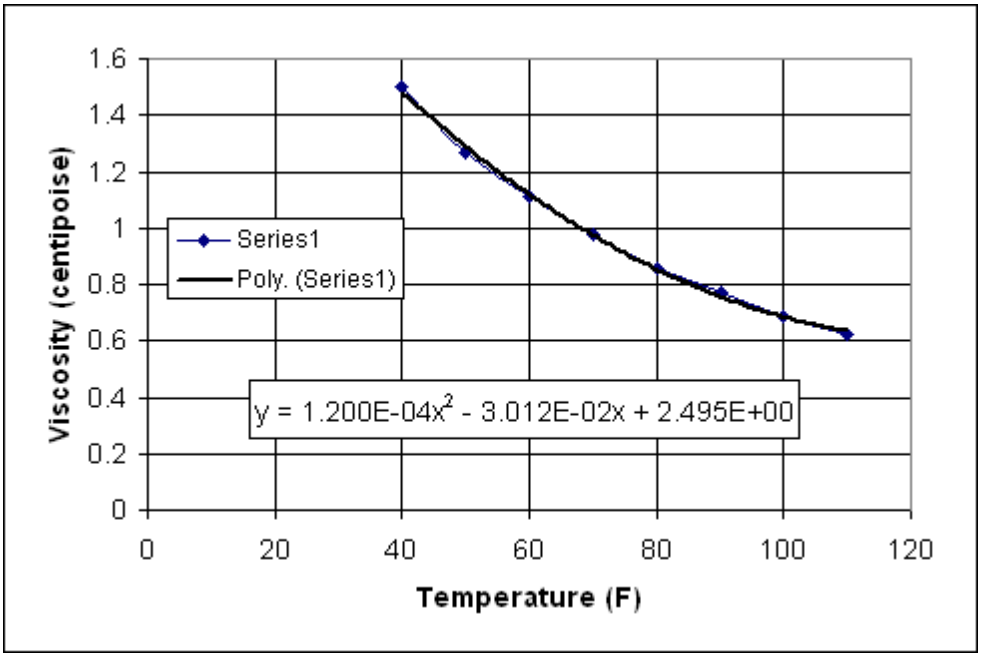
Figure 4 — Reynolds Number factor

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Annex A (informative) — Engineering data

Table A.1 — Properties for water

Temperature		Density	ρ/ρ_o	Absolute Viscosity	Kinematic Viscosity	
(F)	(C)	(lb/ft^3)		(Centipoise)	(Centistokes)	(m^2/sec)
40	4.4	62.426	1.000882	1.500	1.4986	1.4986E-06
50	10.0	62.410	1.000625	1.270	1.2692	1.2692E-06
60	15.6	62.371	1.000000	1.110	1.1100	1.1100E-06
70	21.1	62.305	0.998942	0.976	0.9770	0.9770E-06
80	26.7	62.220	0.997579	0.857	0.8590	0.8590E-06
90	32.2	62.116	0.995912	0.773	0.7761	0.7761E-06
100	37.8	61.996	0.993988	0.685	0.6891	0.6891E-06
110	43.3	61.862	0.991839	0.625	0.6301	0.6301E-06



NOTE 1 — To convert from centipoise to centistokes, divide by ρ/ρ_o where ρ_o = 62.371 lbm/ft^3.

NOTE 2 — To convert from centistokes to m^2/sec, multiply by 1×10^{-6}.

NOTE 3 — In the curve fit above, x is the temperature in degrees Fahrenheit and y is the viscosity in centipoise.

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Table A.2 — Properties of air

Temperature		Absolute Viscosity	
(F)	(C)	(Centipoise, cP)	(Pa*sec)
40	4.4	0.01723	1.723E-05
50	10.0	0.01750	1.750E-05
60	15.6	0.01777	1.777E-05
70	21.1	0.01804	1.804E-05
80	26.7	0.01831	1.831E-05
90	32.2	0.01858	1.858E-05
100	37.8	0.01884	1.884E-05
110	43.3	0.01910	1.910E-05

To convert from centistokes to m^2/sec, multiply by 1 x 10^{-6}.

To calculate the kinematic viscosity, use the equation below.

$\nu = Nv * \mu * T/P$

where ν = the kinematic viscosity,

Nv = a conversion constant depending on the unit used (see below),

μ = the absolute (dynamic) viscosity,

T = the absolute temperature of the air, and

P = the absolute pressure of the air.

Nv	μ	ν	T	p
23.13	cP	cS	R	psi
2.87E-03	Pa*sec	m^2/sec	K	bar
2.87E-01	Pa*sec	m^2/sec	K	kPa

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Table A.3 — Typical values of valve style modifier F_d, liquid pressure recovery factor F_L, and pressure differential ratio factor x_T at full rated travel [1)]

Valve type	Trim type	Flow direction [2)]	F_L	x_T	F_d
Globe, single port	3 V-port plug	Open or close	0.9	0.70	0.48
	4 V-port plug	Open or close	0.9	0.70	0.41
	6 V-port plug	Open or close	0.9	0.70	0.30
	Contoured plug (linear and equal percentage)	Open Close	0.9 0.8	0.72 0.55	0.46 1.00
	60 equal diameter hole drilled cage	Outward [3)] or inward [3)]	0.9	0.68	0.13
	120 equal diameter hole drilled cage	Outward [3)] or inward [3)]	0.9	0.68	0.09
	Characterized cage, 4-port	Outward [3)] Inward [3)]	0.9 0.85	0.75 0.70	0.41 0.41
Globe, double port	Ported plug	Inlet between seats	0.9	0.75	0.28
	Contoured plug	Either direction	0.85	0.70	0.32
Globe, angle	Contoured plug (linear and equal percentage)	Open Close	0.9 0.8	0.72 0.65	0.46 1.00
	Characterized cage, 4-port	Outward [3)] Inward [3)]	0.9 0.85	0.65 0.60	0.41 0.41
	Venturi	Close	0.5	0.20	1.00
Globe, small flow trim	V-notch	Open	0.98	0.84	0.70
	Flat seat (short travel)	Close	0.85	0.70	0.30
	Tapered needle	Open	0.95	0.84	$N_{19}\frac{(CF_L)^{0.5}}{D_o}$
Rotary	Eccentric spherical plug	Open Close	0.85 0.68	0.60 0.40	0.42 0.42
	Eccentric conical plug	Open Close	0.77 0.79	0.54 0.55	0.44 0.44
Butterfly (centered shaft)	Swing-through (70°)	Either	0.62	0.35	0.57
	Swing-through (60°)	Either	0.70	0.42	0.50
	Fluted vane (70°)	Either	0.67	0.38	0.30
High Performance Butterfly (eccentric shaft)	Offset seat (70°)	Either	0.67	0.35	0.57
Ball	Full bore (70°)	Either	0.74	0.42	0.99
	Segmented ball	Either	0.60	0.30	0.98

NOTE 1 — These values are typical only; actual values shall be stated by the valve manufacturer.

NOTE 2 — Flow tends to open or close the valve, i.e. push the closure device (plug, ball, or disc) away from or towards the seat.

NOTE 3 — Outward means flow from center of cage to outside, and inward means flow from outside of cage to center.

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Units conversions and variable changes

The following portion of Annex A is adapted from ANSI/ISA-75.01.01 (IEC 60534-2-1 Mod)-2007. The material is included for ease of referencing.

Example:

The pressure is known to be 23.00psi, but is required in units of bar.

23.00psi x 0.0689476 = 1.586bar

Pressure

$1\ \text{bar} = 14.5037738\ \text{psi}$ $\qquad$ $1\ \text{psi} = 0.0689476\ \text{bar}$

$1\ \frac{\text{kgf}}{\text{cm}^2} = 14.2233433\ \text{psi}$ $\qquad$ $1\ \text{psi} = 0.070307\ \frac{\text{kgf}}{\text{cm}^2}$

$1\ \text{kPa} = 0.1450377\ \text{psi}$ $\qquad$ $1\ \text{psi} = 6.8947573\ \text{kPa}$

Liquid Flow Rate

$1\ \frac{\text{bbl}}{\text{day}} = 42\ \frac{\text{gal}}{\text{day}}$ $\qquad$ $1\ \frac{\text{gal}}{\text{day}} = 0.0238095\ \frac{\text{bbl}}{\text{day}}$

$1\ \frac{\text{m}^3}{\text{hr}} = 4.4028673\ \frac{\text{gal}}{\text{min}}$ $\qquad$ $1\ \frac{\text{gal}}{\text{min}} = 0.2271247\ \frac{\text{m}^3}{\text{hr}}$

$1\ \frac{\text{m}^3}{\text{hr}} = 150.9554512\ \frac{\text{bbl}}{\text{day}}$ $\qquad$ $1\ \frac{\text{bbl}}{\text{day}} = 6.6244709 \times 10^{-3}\ \frac{\text{m}^3}{\text{hr}}$

$1\ \frac{\text{kg}}{\text{hr}} = 2.2046226\ \frac{\text{lbm}}{\text{hr}}$ $\qquad$ $1\ \frac{\text{lbm}}{\text{hr}} = 0.4535924\ \frac{\text{kg}}{\text{hr}}$

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Gas Flow Rate

$$1\frac{m^3}{hr} = 35.3146667\ \frac{ft^3}{hr} \qquad \frac{ft^3}{hr} = 0.0283168\ \frac{m^3}{hr}$$

$$q_{actual} = q_{STD}\ \frac{TZ}{P}\ \frac{P_s}{T_s \bullet 1.0}$$

where q_{actual} is the actual volumetric flowrate at flowing conditions,
q_{STD} is the volumetric flowrate at standard conditions,
T is the actual absolute temperature,
T_s is the absolute temperature at standard conditions (60°F, 288.6K),
P is the actual absolute pressure,
P_s is the standard absolute pressure (14.70 psi, 1.01325 bar), and
Z is the compressibility at actual conditions (Z at standard conditions is assumed to equal 1.0, which is represented by 1.0 in the equation above).
Note that the units used for T_s must be the same as the units used for T and the units used for P_s must be the same as the units used for P.

$$\rho = \frac{MP}{N_{U1}TZ}$$

where ρ is the gas density
M is the molecular weight
N_{U1} is a constant whose numeric value is indicated in Table A.4,
P is the absolute pressure,
T is the absolute temperature, and
Z is the gas compressibility.

Note that the absolute temperature, in K equals 273.15 plus the temperature in degrees C and the absolute temperature in degrees R equals 459.67 plus the temperature in degrees F.

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Table A.4 - Values of N_{U1}

N_{U1}	ρ	P	T
8.3126	kg/m^3	kPa	K
0.6698	kg/m^3	psi	^oR
0.084765	kg/m^3	kg_f/cm^2	K
0.083126	kg/m^3	bar	K
133.155	lb_m/ft^3	kPa	K
10.729	lb_m/ft^3	psi	^oR
1.3578	lb_m/ft^3	kg_f/cm^2	K
1.3316	lb_m/ft^3	bar	K

Absolute (or Dynamic) Viscosity

$1\ \text{cP} = 0.01\ \text{poise}$ $\qquad 1\ \text{poise} = 100\ \text{cP}$

$1\ \text{cP} = 2.0885434 \times 10^{-5}\ \frac{\text{lbf}\cdot\text{s}}{\text{ft}^2}$ $\qquad 1\ \frac{\text{lbf}\cdot\text{s}}{\text{ft}^2} = 4.7880259 \times 10^{4}\ \text{cP}$

$1\ \text{cP} = 6.7196898 \times 10^{-4}\ \frac{\text{lbm}}{\text{ft}\cdot\text{s}}$ $\qquad 1\ \frac{\text{lbm}}{\text{ft}\cdot\text{s}} = 1.4881639 \times 10^{3}\ \text{cP}$

$1\ \text{cP} = 2.0885434 \times 10^{-5}\ \frac{\text{slug}}{\text{ft}\cdot\text{s}}$ $\qquad 1\ \frac{\text{slug}}{\text{ft}\cdot\text{s}} = 4.7880259 \times 10^{4}\ \text{cP}$

$1\ \text{cP} = 1 \times 10^{-3}\ \text{Pa}\cdot\text{s}$ $\qquad 1\ \text{Pa s} = 1 \times 10^{3}\ \text{cP}$

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Kinematic Viscosity

$$1\text{cS} = 1 \times 10^{-6} \frac{\text{m}^2}{\text{s}} \qquad 1 \frac{\text{m}^2}{\text{s}} = 1 \times 10^{6} \text{ cS}$$

$$1\text{cS} = 1.076391 \times 10^{-5} \frac{\text{ft}^2}{\text{s}} \qquad 1 \frac{\text{ft}^2}{\text{s}} = 9.290304 \times 10^{4} \text{ cS}$$

$$\nu = N_{U2} \frac{\mu}{\rho}$$

where v is the kinematic fluid viscosity,

N_{U2} is a constant whose numeric value is indicated in Table A.5,

μ is the fluid absolute (or dynamic) fluid viscosity, and

ρ is the fluid density.

Table A.5 - Values of N_{U2}

N_{U2}	*v*	μ	ρ
1000	cS	cP	kg/m^3
1.4882×10^{6}	cS	$lb_m/ft\text{-}s$	kg/m^3
1	m^2/s	Pa-s	kg/m^3
62.42796	cS	cP	lb_m/ft^3
1	ft^2/s	$lb_m/ft\text{-}s$	lb_m/ft^3

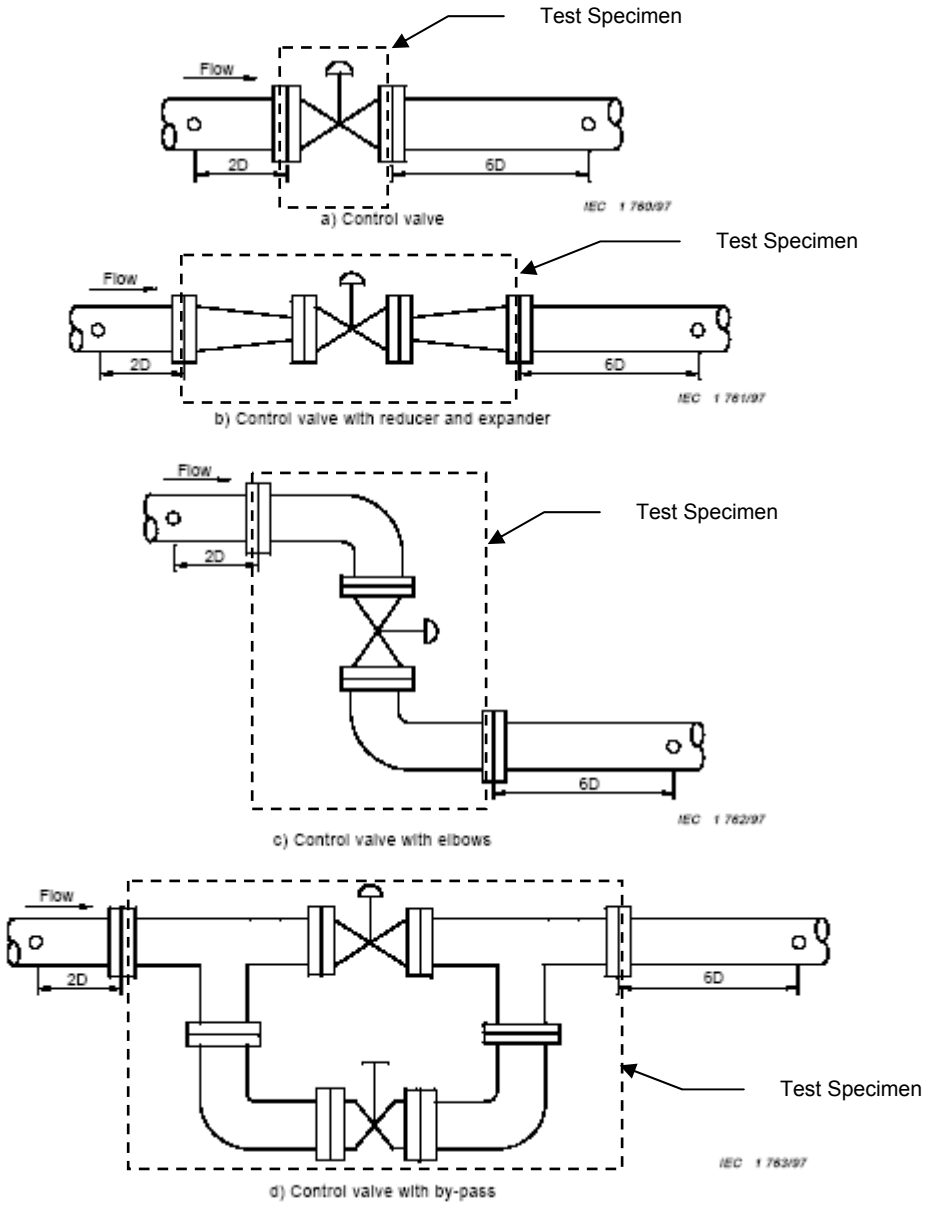
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Annex B (informative) —Tap location and setup diagrams for common field installations

Following are examples of test specimens depicting common field installations indicating appropriate pressure tap locations.



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It should be noted that all procedures and data reduction equations presented throughout this document assume that both the upstream pressure and downstream pressure tap locations fall in the same horizontal plane, i.e., elevation change between the tap locations is not included in the data reduction.

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Annex C (informative) — Derivation of the valve style modifier, F_d

Annex C is extracted from ANSI/ISA-75.01.01 (IEC 60534-2-1 Mod)-2007. The material is duplicated within this standard for ease of referencing.

All variables in this annex have been defined in this part except for the following:

A_o area of vena contracta of a single flow passage, millimeters squared;

d_H hydraulic diameter of a single flow passage, millimeters;

d_i inside diameter of annular flow passage (see Figure C.1), millimeters;

d_o equivalent circular diameter of the total flow area, millimeters;

D_o diameter of seat orifice (see Figures C.1 and C.2), millimeters;

l_w wetted perimeter of a single flow passage, millimeters;

N_o number of independent and identical flow passages of a trim, dimensionless;

α angular rotation of closure member (see Figure C.2), degrees;

β maximum angular rotation of closure member (see Figure C.2), degrees;

ζ_{B1} velocity of approach factor, dimensionless; and

μ discharge coefficient, dimensionless.

The valve style modifier F_d, defined as the ratio d_H / d_o at rated travel and where $C_i / d^2 > 0.016\ N_{18}$, may be derived from flow tests using the following equation:

(Eq. C.1)
$$F_d = \frac{N_{26} \nu F_L^{\,2} F_R^{\,2} \left(C / d^2\right)^2 \sqrt{C F_L}}{Q \left(\frac{F_L^{\,2} C^2}{N_2 D^4} + 1 \right)^{1/4}}$$

For valves having $C_i / d^2 \leq 0.016\ N_{18}$, F_d is calculated as follows:

(Eq. C.2)
$$F_d = \frac{N_{31} \nu F_L^{\,2} F_R^{\,2} \sqrt{C F_L}}{Q \left[1 + N_{32} \left(\frac{C}{d^2} \right)^{2/3} \right]}$$

NOTE — Values for N_{26} and N_{32} are listed in Table C.1.

The test for determining F_d is covered in IEC 60534-2-3.

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Alternatively, F_d can be calculated by the following equation:

(Eq. C.3) $$F_d = \frac{d_H}{d_o}$$

The hydraulic diameter d_H of a single flow passage is determined as follows:

(Eq. C.4) $$d_H = \frac{4A_o}{l_w}$$

The equivalent circular diameter d_o of the total flow area is given by the following equation:

(Eq. C.5) $$d_o = \sqrt{\frac{4N_o A_o}{\pi}}$$

F_d may be estimated with sufficient accuracy from dimensions given in manufacturers' drawings.

The valve style modifier for a single-seated, parabolic valve plug (flow tending to open) (see Figure C.1) may be calculated from Equation C.3.

From Darcy's equation, the area A_o is calculated from the following equation:

(Eq. C.6) $$A_o = \frac{N_{23} C F_L}{N_o}$$

NOTE — Values for N_{23} are listed in Table C.1.

Therefore, since $N_o = 1$,

(Eq. C.7) $$d_o = \sqrt{\frac{4A_o}{\pi}}$$

$$= \sqrt{\frac{4N_{23} C F_L}{\pi}}$$

(Eq. C.8) $$d_H = \frac{4A_o}{l_w}$$

$$= \frac{4N_{23} C F_L}{\pi(D_o + d_i)}$$

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From above,

(Eq. C.3) $$F_d = \frac{d_H}{d_o}$$

$$= \frac{\left[\frac{4N_{23}CF_L}{\pi\left(D_o + d_i\right)}\right]}{\sqrt{\frac{4N_{23}CF_L}{\pi}}}$$

(Eq. C.9) $$= \frac{1.13\sqrt{N_{23}CF_L}}{D_o + d_i}$$

where d_i varies with the flow coefficient. The diameter d_i is assumed to be equal to zero when $N_{23}CF_L = D_o^2$. At low C values, $d_i \approx D_o$; therefore,

(Eq C.10) $$d_i = D_o - \frac{N_{23}CF_L}{D_o}$$

(Eq. C.11) $$F_d = \frac{1.13\sqrt{N_{23}CF_L}}{2D_o - \frac{N_{23}CF_L}{D_o}}$$

The maximum F_d is 1.0.

For swing-through butterfly valves, see Figure C.2.

The effective orifice diameter is assumed to be the hydraulic diameter of one of the two jets emanating from the flow areas between the disk and valve body bore; hence $N_o = 2$.

The flow coefficient C at choked or sonic flow conditions is given as

(Eq. C.12) $$N_{23}CF_L = \frac{0.125\pi D_o^{\,2}\left(\mu_1 + \mu_2\right)\left(\frac{1-\sin\alpha}{\sin\beta}\right)}{\zeta_{B1}}$$

Assuming the velocity of approach factor $\zeta_{B1} = 1$, making $\mu_1 = 0.7$ and $\mu_2 = 0.7$, and substituting Equation C.6 into Equation C.12 yields Equation C.13.

(Eq. C.13) $$A_o = \frac{0.55D_o^{\,2}\left(\frac{1-\sin\alpha}{\sin\beta}\right)}{N_o}$$

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and since β = 90° for swing-through butterfly valves,

(Eq. C.14) $$A_o = \frac{0.55 D_o^{\,2}(1-\sin\alpha)}{N_o}$$

However, since there are two equal flow areas in parallel,

(Eq. C.15) $$A_o = 0.275 D_o^{\,2}(1-\sin\alpha)$$

and $$d_o = \sqrt{\frac{4A_oN_o}{\pi}}$$

(Eq. C.16) $$= 0.837 D_o\sqrt{1-\sin\alpha}$$

$$d_H = \frac{4A_o}{0.59\pi D_o}$$

(Eq. C.17) $$= 0.59 D_o(1-\sin\alpha)$$

NOTE — 0.59 π D_o is taken as the wetted perimeter l_w of each semi-circle allowing for jet contraction and hub.

(Eq. C.3) $$F_d = \frac{d_H}{d_o}$$

which results in

(Eq. C.18) $$F_d = 0.7\sqrt{1-\sin\alpha}$$

Table C.1 — Numerical constant *N*

Constant	Flow coefficient C		Formulae unit		
	K_v	C_v	Q	d	ν
N_{23}	1.96×10^{1}	1.70×10^{1} 2.63×10^{-2}	– –	mm in	– –
N_{26}	1.28×10^{7}	9.00×10^{6} 9.52×10^{-5}	m^3C_v /h gpm	mm in	m^2/s cS
N_{31}	2.1×10^{4}	1.9×10^{4} 8.37×10^{-2}	m^3/h gpm	mm in	m^2/s cS
NOTE — Use of the numerical constant provided in this table together with the practical metric units specified in the table will yield flow coefficients in the units in which they are defined.					

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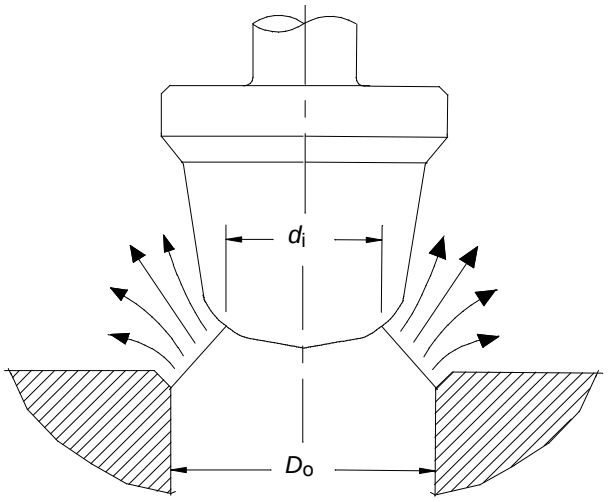


Figure C.1 — Single seated, parabolic plug (flow tending to open)

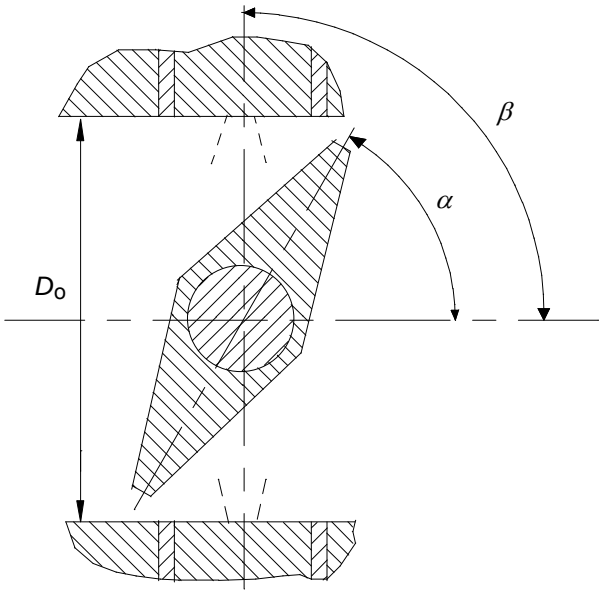


Figure C.2 — Swing-through butterfly valve

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Annex D (informative) — Laminar flow test discussion and bibliography

The flow coefficient, C (C_v, K_v), is defined and normally measured under fully turbulent conditions. Establishing appropriate flow conditions for measuring the flow coefficient of very low flow valve trims can be difficult, however, especially when the coefficient on the order of 0.01 or less. While there is agreement that nonturbulent flow for such valves can be adequately predicted a universally accepted approach within the industry is currently lacking. It follows that there is diversity in the approach to measuring the coefficients defined in this standard.

In order of preference:

Turbulent flow with water

Turbulent flow with compressible media

Laminar flow with compressible media

In addition to ANSI/ISA-75.01.01-2000, the following bibliography is offered for the interested reader:

Stiles, G. F. 1967, “Liquid Viscosity Effects on Control Valve Sizing,” Technical Monogram TM17, Fisher Controls International, Marshalltown, IA

McCutcheon, E. D, 1974, “A Reynolds Number for Control Valves,” Symposium on Flow, its Measurement and Control in Science and Industry, Vol 1, Part 3.

George, J. A., 1989, “Sizing and Selection of Low Flow Control Valves,” InTech, November 1989.

Baumann, H. D., 1991, “Viscosity Flow Correction for Small Control Valve Trim,” Transactions of the ASME, Vol. 113

Baumann, H. D., 1993, “A Unifying Method for Sizing Throttling Valves Under Laminar Flow or Transitional Flow Conditions,” Transactions of the ASME, Vol. 115

Kitterredge, C. P. and Rowley, D. S., 1957, “Resistance Coefficients for Laminar and Turbulent Flow through One Half Inch Valves and Fittings,’” Transactions of ASME, Vol 79, pp. 1759-1766

Crane Technical Paper 410, “Flow of Fluids through Valves, Fittings and Pipe,” 1976, pp. 3-4

Kiesbauer, J., 1995, “Calculation of the Flow Behavior of Micro Control Valves,” SAMSON AG

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Annex E (informative) — Long form F_L test procedure

The following is a description of an alternate method of evaluating the Liquid Pressure Recovery Factor, F_L. Referred to herein as the "long form" method, it expands the data set upon which the F_L value is determined. The advantage of this method is that it renders a more comprehensive characterization of flow over the full domain of pressure drop ratio. These results can reveal important information regarding the behavior of the valve that may not be apparent in the abbreviated "standard" version.

E.1 Test procedure

E.1.1 The test specimen shall be installed in a test system as prescribed by Clause 4 of this standard. The test shall be conducted utilizing an incompressible test fluid as specified in 5.1. All data shall be collected and recorded per 6.2.3.

E.1.2 The valve travel shall be set to the desired value and the maximum flow rate and pressure different established in accord with the procedure described in 6.2.2 of this standard.

E.1.3 Additional test pressure differentials shall be established such that 10-15 data points exist uniformly over the full test pressure differential range (zero to the maximum differential established in E.2). Beginning at the choked flow condition, steady state flow shall be established at each pressure differential in decreasing order and data recorded.

E.1.4 If the test procedure is disrupted for any reason, the initial test pressure differential on resuming testing shall be established by exceeding the target value by a minimum of 10% and decreasing the pressure drop to the desired value.

E.1.5 The preliminary data shall be reduced per E.2 below and additional test runs conducted as needed to fully define the flow profile of the test specimen. In particular, additional data points should be collect at inflection points on the resulting curve, or near regions of high curvature.

E.2 Graphical data reduction

E.2.1 The value of F_L is established by determining the common pressure differential solution to the incompressible volumetric flow equation,

(Eq. E.1) $$Q = C_V \sqrt{\frac{\Delta P}{G_f}}$$

and the incompressible choked flow equation,

(Eq. E.2) $$Q = Q_c$$

This value is substituted into the defining F_L expression:

(Eq. E.3) $$F_L = \sqrt{\frac{\Delta P}{P_1 - F_F P_v}}$$

to yield

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(Eq. E.4)

$$F_L = \left(\frac{Q_c}{C_V}\right)\sqrt{\frac{G_f}{P_1 - F_F P_v}}$$

The mechanics of analyzing the flow data is centered on establishing representative values for the choked flow rate, Q_c, and incompressible flow coefficient, C_v, values in equation E.4. The procedure presented herein is graphically based to illustrate the principals underlying data reduction. It is recognized that a variety of regression schemas can be employed to automate the procedure.

E.2 The results of the testing should be imaged by plotting flow rate, Q, vs. the square root of the applied pressure differential as shown in Figure E.1.

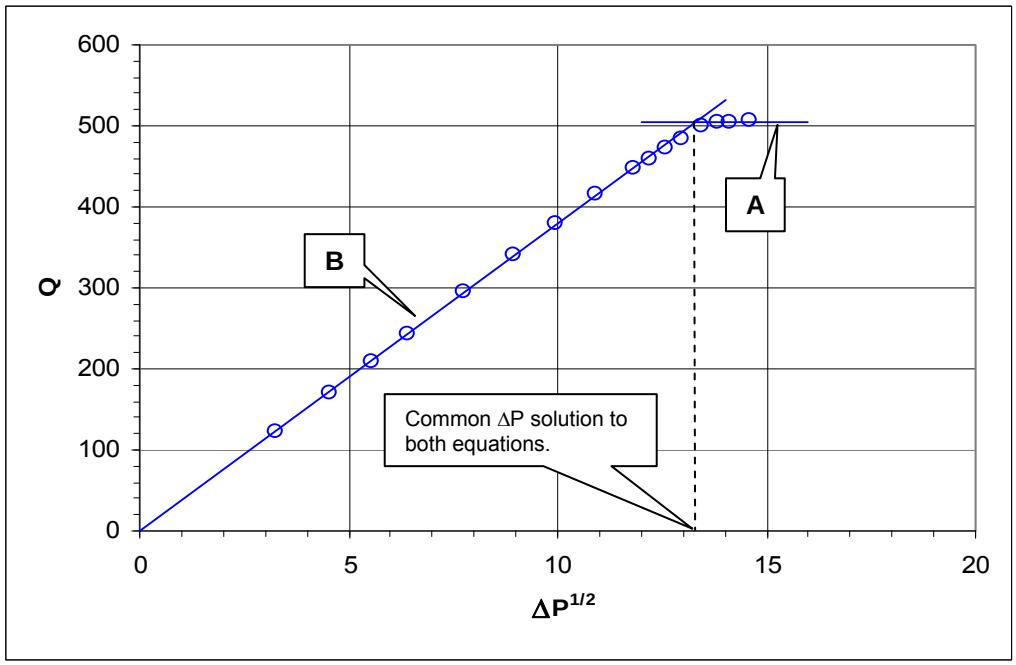


Figure E.1 — Typical flow results

E.3 A straight line representative of the choked flow rate should be established on the basis of the data and the value of Q_c noted (line A, Figure E.1).

E.4 A second straight line representative of the incompressible portion of the flow curve should be established (line B, Figure E.1). The line should pass through the origin of the graph and represent the data throughout the incompressible region. The slope of this line corresponds to the incompressible flow coefficient, C_v. The value of C_v as determined in 7.1 may alternatively be used to establish the slope of the curve.

E.5 The value of Q_c and C_v resulting from the graphical analysis is used in conjunction with equation E.4 to compute the value of F_L.

NOTE — The value of F_L and the value of C_v used to evaluate F_L constitute a matched pair of values. Published data values of F_L should be consistent with published values of C_v.

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Annex F (informative) — Calculation of F_P to help determine if pipe/valve port diameters are adequately matched

NOTE — The term "port" in the context of the following discussion refers to "the opening of a valve's inlet or outlet passageways" per ANSI/ISA-75.05.01-2000 (R2005), 3.120 (2).

As mentioned in 4.3, the valve and pipe port diameters shall be matched closely enough to not introduce significant errors in the calculations. This, of course, assumes that the intent is the most common one where the upstream and downstream piping is the same size as the valve. If the characteristics of a particular valve/pipe configuration where some or all of the piping is not the same size as the valve are desired, one of the goals would be the calculation of a pipe geometry factor, F_P, as described in 8.4; otherwise the upstream and downstream piping should match. Matching pipe and valve port inside diameters is often not difficult with ordinary pipe sizes and schedules but in some cases, such as the testing of a very high pressure valve with small port inside diameters, special piping may be required. This standard specifies a method for determining the suitability of pipe inside diameters. Subclause 4.3 specifies that the estimated piping geometry factor, calculated using formulas given in ANSI/ISA-75.01.01 (IEC 60534-2-1 Mod)-2007 and repeated below for convenience, must be within the range 0.99 to 1.02, i.e. $0.99 \le F_P \le 1.01$. F_P is calculated from

(Eq. F.1)
$$F_P = \frac{1}{\sqrt{1 + \frac{\sum \zeta}{N_2}\left(\frac{C}{d^2}\right)^2}}$$

where $\sum \zeta$ is the sum of upstream and downstream Bernoulli coefficients and loss coefficients. They are calculated using Eqs. F.2 through F.6 below and are adaptations of Eqs. 20 through 24 of ANSI/ISA-75.01.01.

(Eq. F.2)
$$\sum \zeta = \zeta_1 + \zeta_2 + \zeta_{B1} - \zeta_{B2}$$

(Eq. F.3)
$$\zeta_{B1} = 1 - \left(\frac{d}{D_1}\right)^4$$

(Eq. F.4)
$$\zeta_{B2} = 1 - \left(\frac{d}{D_2}\right)^4$$

(Eq. F.5)
$$\zeta_1 = 0.5\left[1 - \left(\frac{d}{D_1}\right)^2\right]^2$$

(Eq. F.6)
$$\zeta_2 = 1\left[1 - \left(\frac{d}{D_2}\right)^2\right]^2$$

The subscripts 1 or 2 indicate upstream or downstream factors respectively. Note that for the purpose of determining F_P here, the valve diameter, d, must be the actual inside diameter of the associated valve port and not the valve nominal diameter. The pipe diameters D_1 and D_2 are pipe inside diameters.

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Two cases are probably most common in testing according to this standard—(1) the upstream and downstream pipe inside diameters are the same size and larger that the valve port inside diameters and (2) the upstream pipe inside diameter is the same size as the valve inside diameter but the downstream pipe inside diameter is larger. Tables F1 and F2 below, tabulate F_P factors for those two cases as a function of the ratios d/D and $\frac{C}{d^2\sqrt{N_2}}$. Note that the large number of digits displayed were included to help verify hand or computer calculations and not to imply high accuracy.

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Table F1 — Tabulated values of F_P if upstream and downstream pipe the same size

$\frac{C}{d^2\sqrt{N_2}}$	d/D_1 or d/D_2				
	1	**0.95**	**0.9**	**0.85**	**0.8**
0.05	1	0.999982	0.999932	0.999856	0.999757
0.1	1	0.999929	0.999729	0.999423	0.999029
0.2	1	0.999715	0.998919	0.997698	0.996135
0.3	1	0.999359	0.997572	0.994842	0.991365
0.4	1	0.998861	0.995696	0.990885	0.984802
0.5	1	0.998222	0.993299	0.985867	0.976551
0.6	1	0.997443	0.990393	0.979835	0.966744
0.7	1	0.996525	0.986992	0.972848	0.955525
0.8	1	0.995468	0.98311	0.964968	0.943054
0.9	1	0.994275	0.978765	0.956265	0.929493
1	1	0.992946	0.973977	0.946811	0.915008

Table F2 — Tabulated values of F_P if downstream pipe larger than valve

$\frac{C}{d^2\sqrt{N_2}}$	d/D_2				
	1	**0.95**	**0.9**	**0.85**	**0.8**
0.05	1	1.00022	1.000385	1.000502	1.000576
0.1	1	1.000881	1.001543	1.002011	1.002312
0.2	1	1.003538	1.006213	1.008118	1.009345
0.3	1	1.008015	1.014146	1.018548	1.021404
0.4	1	1.014383	1.025573	1.03371	1.039036
0.5	1	1.022752	1.040848	1.054237	1.063108
0.6	1	1.033267	1.060479	1.081069	1.094934
0.7	1	1.046122	1.085177	1.115585	1.136505
0.8	1	1.061569	1.115938	1.15984	1.190908
0.9	1	1.07993	1.154176	1.216981	1.263142
1	1	1.101623	1.201944	1.292058	1.361837

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Annex G (informative)— Bibliography

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60534-1 Part 1: Control Valve Terminology and General Considerations, 2005

IEC 60534-2-1 Part 2-1: Flow Capacity; Sizing Equations for Fluid Flow Under Installed Conditions, 1998

IEC 60534-2-3 Part 2-3: Flow Capacity - Test Procedures, 1997

Available from: **ANSI**
25 West 43rd Street
Fourth Floor
New York, NY 10036

ISA

ANSI/ISA-75.01.01 (IEC 60534-2-1 Mod)-2007, Flow Equations for Sizing Control Valves

ANSI/ISA-75.05.01-2000 (R2005), Control Valve Terminology

Available from: **ISA**
67 Alexander Drive
PO Box 12277
Research Triangle Park, NC 27709
Tel: (919) 990-9200

ASME

ASME Performance Test Code PTC 19.5-2004, "Applications."

ASME Performance Test Code PTC 19.5-1972, "Applications. Part II of Fluid Meters, Interim Supplement on Instruments and Apparatus."

ASME *Fluid Meters* for additional guidelines for line length

Available from: **ASME**
ASME International
Three Park Avenue
New York, NY 10016-5990
Tel: (800) 843-2763

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Cunningham, R.G., “Orifice Meters with Supercritical Compressible Flow,” ASME Transactions 73, pp. 625-638, July 1951.

Driskell, L. R., “New Approach to Control Valve Sizing,” Hydrocarbon Processing, pp. 131-134, July 1969.

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